

Mean-Value Coordinates on Integer Factor Lattices: A General Conic Theorem, the Divisor Problem, and a Correspondence with $SU(2)$

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Abstract

For a pair of positive integer factors (x, y) , set $X = (x - y)/2$, $Y = \sqrt{xy}$, $T = (x + y)/2$ — the semi-difference, geometric mean, and arithmetic mean. These satisfy the AM–GM identity $X^2 + Y^2 = T^2$, placing every factor pair on a cone. We prove a single classification theorem that organizes every straight line through the multiplication table: for a line $ax + by = c$, the geometric mean Y is a quadratic function of x , and the resulting curve on the cone is an ellipse when $ab > 0$ (with the classical anti-diagonal circle as the symmetric special case $a = b$), a parabola when $ab = 0$ (rows and columns), and — distinct from the constant-product hyperbolas studied separately — a second hyperbola family when $ab < 0$. We verify each case exactly, including a genuine ellipse (confirmed via the conic discriminant test), and identify the classical mechanism behind each: a cutting plane’s relationship to the cone’s axis and generators. We then show this coordinate system reorganizes the classical Dirichlet divisor summatory function $D(n)$ via nested gnomonic shells, produces a numerical match between the multiplication table and $SU(2)$ ladder-operator matrix elements — sharpened by the exact algebraic fact that the same substitution turns the cone identity itself into the $\mathfrak{su}(2)$ Casimir relation — and identifies constant-product curves $xy = N$ with invariant hyperbolas of a Minkowski norm — stating precisely, in each case, what is and is not established. We then prove that no nontrivial Lorentz boost preserves the integer divisor lattice of $xy = N$, tracing the reason to the classical Pell dichotomy for binary quadratic forms, and prove that among moduli with a Klein four-group of units, only 12 admits a bijection between its quadratic residues and its totatives — using this to give a precise bridge to the $SU(2)$ correspondence, and showing that mod 12’s forced real Dirichlet characters turn a numerical observation about the divisor problem’s error term $\Delta(n)$ into a proven fact about its mean behavior.

1 Introduction

For a factor pair (x, y) , set $X = (x - y)/2$, $Y = \sqrt{xy}$, $T = (x + y)/2$. These satisfy $X^2 + Y^2 = T^2$, so every factor pair sits on a cone, and multiplication tables are usually read along rows and columns when the more informative structure is along the lines that cut through them. This note is organized around one question: what curve does a general line $ax + by = c$ through the table trace on that cone? Section 2 sets up the coordinate system and its first, classical example, the circle from the anti-diagonal $x + y = K$. Section 3 answers the general question completely: every line’s type — ellipse, parabola, or a second hyperbola family — is fixed by the sign of ab , with circles and the row/column parabolas as special cases of one theorem rather than three separate facts. Section 4 studies a different, non-linear curve family, the constant-product hyperbolas $xy = N$, and its identification with an invariant of a $(2 + 1)$ -dimensional Minkowski norm. Section 5 shows what

this coordinate system buys classically: a re-derivation of the divisor summatory function $D(n)$ via nested gnomonic shells. Section 6 reports an exact numerical correspondence between the table and $SU(2)$ ladder-operator matrix elements, stated carefully as a coordinate relabeling rather than a structural isomorphism. Section 7 proves that no nontrivial Lorentz boost preserves the integer divisor lattice of $xy = N$, and Section 8 proves that modulus 12 is the unique Klein-four modulus admitting a bijection between quadratic residues and totatives, with consequences for $SU(2)$ and for the divisor problem's error term.

2 Mean-Value Coordinates and the Conic Identity

Define, for a factor pair (x, y) with $x, y > 0$,

$$X = \frac{x - y}{2}, \quad Y = \sqrt{xy}, \quad T = \frac{x + y}{2}. \quad (2.1)$$

These are respectively the semi-difference, the geometric mean, and the arithmetic mean of x and y . They satisfy the AM–GM identity

$$X^2 + Y^2 = T^2, \quad (2.2)$$

which is elementary — it is the statement $\left(\frac{x-y}{2}\right)^2 + xy = \left(\frac{x+y}{2}\right)^2$, verified by direct expansion — but it has a convenient geometric reading. Fixing the anti-diagonal $x + y = K$ fixes $T = K/2$, so (2.2) restricts to the circle $X^2 + Y^2 = (K/2)^2$. As x ranges over $1, \dots, K-1$ with $y = K - x$, the point (X, Y) traces $K-1$ discrete points on that circle, with the diagonal point $x = y = K/2$ (when K is even) sitting at the apex $X = 0, Y = K/2$. Stacking these circles as K varies produces the double-napped cone $X^2 + Y^2 = T^2$ in (X, Y, T) -space. This is a faithful, if elementary, geometric picture of the multiplication table's anti-diagonal structure, and it is the coordinate system used throughout this note.

The identity (2.2) is also exactly the classical geometric mean theorem for a right triangle: if the altitude from the right angle splits the hypotenuse into two segments of length p and q , that altitude has length $\sqrt{pq}$, and the hypotenuse itself has length $p + q$. Setting $p = x, q = y$ recovers Y as the altitude and $T = (p + q)/2$ as the semi-hypotenuse; X is then simply the signed distance from the midpoint of the hypotenuse to the foot of the altitude. So the coordinates (X, Y, T) are not a bespoke construction — they are the standard right-triangle picture underlying the geometric mean, applied to each factor pair in turn.

3 The General Conic Theorem: Every Line Through the Table

The anti-diagonal $x + y = K$ is one line through the multiplication table among infinitely many, and Section 2 showed it produces a circle. This section answers the natural question completely: what does an arbitrary line $ax + by = c$ produce, and the answer turns out to organize circles, ellipses, parabolas, and a second family of hyperbolas under a single classification.

3.1 Statement and proof

Theorem 3.1. *Let a, b be real numbers, not both zero, and c a real number, and consider the line $ax + by = c$ intersecting the multiplication table. On the cone $X^2 + Y^2 = T^2$ of Section 2, this line traces:*

- *an ellipse if $ab > 0$, with the circle of Section 2 as the symmetric special case $a = b$;*

- a parabola if $ab = 0$ — the rows ($b = 0$) and columns ($a = 0$) of the table;
- a hyperbola if $ab < 0$ — a family distinct from the constant-product hyperbolas of Section 4.

Proof. For $b \neq 0$, solve $ax + by = c$ for $y = (c - ax)/b$ and substitute into $Y^2 = xy$:

$$Y^2 = \frac{c}{b}x - \frac{a}{b}x^2, \quad (3.1)$$

a quadratic function of x with leading coefficient $-a/b$. When $ab > 0$ this coefficient is negative, so Y^2 rises from 0, reaches a maximum, and returns to 0 as x ranges over its real domain — the real locus is bounded. A bounded, non-degenerate real conic is an ellipse; the case $a = b$ is the anti-diagonal $x + y = c/a$, whose ellipse degenerates by symmetry to the circle of Section 2. When $ab < 0$ the coefficient $-a/b$ is positive, so Y^2 grows without bound and the locus is an unbounded hyperbola. The case $b = 0$ (or symmetrically $a = 0$) is the row $x = c/a$ (or column $y = c/b$), treated directly in Section 3.2. $\square$

We verified the ellipse case computationally as well, both as a sanity check on the boundedness argument and as an independent confirmation via the standard discriminant test for a general conic $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. For the line $2x + 5y = 40$ (so $ab = 10 > 0$), fitting this conic to the resulting curve — expressed in the plane's own two natural coordinates — gives discriminant $B^2 - 4AC \approx -0.0065$, confirming an ellipse; the same fit at $a = b$ (the anti-diagonal) gives the circle exactly, and at $b \rightarrow 0$ the fit degenerates toward the unbounded parabola of Section 3.2.

3.2 The parabola case: rows and columns

Theorem 3.2. *For a row $x = c$ of the multiplication table ($y = 1, 2, 3, \dots$), the coordinates satisfy $X + T = c$ and $Y^2 = c(T - X)$; for a column $y = c$, they satisfy $T - X = c$ and $Y^2 = c(X + T)$. Both are parabolas.*

Proof. Substituting $x = c$ into $X = (x - y)/2$ and $T = (x + y)/2$ gives $X + T = c$ identically, independent of y ; writing $W = T - X$, direct substitution gives $W = y$, and $Y^2 = xy = cy = cW$. The column case is symmetric, substituting $y = c$ instead. $\square$

Concretely: the row $x = 1$ (giving $1, 2, 3, 4, \dots$) satisfies $X + T = 1$ throughout, with $Y^2 = y$ taking the values $1, 2, 3, 4, \dots$; the row $x = 2$ (giving $2, 4, 6, 8, \dots$) satisfies $X + T = 2$, with $Y^2 = 2y$ taking the values $2, 4, 6, 8, \dots$; the row $x = 3$ (giving $3, 6, 9, 12, \dots$) satisfies $X + T = 3$, with $Y^2 = 3y$ taking the values $3, 6, 9, 12, \dots$, and so on for every row $x = c$.

This is the classical construction of a parabola as a conic section, not a coincidence of algebra. The plane $X + T = c$ has normal vector $(1, 0, 1)$. The cone $X^2 + Y^2 = T^2$ has two special directions running along its surface through the origin, its generators; one of them is $(-1, 0, 1)$, which lies on the cone since $(-1)^2 + 0^2 = 1^2$. Since $(1, 0, 1) \cdot (-1, 0, 1) = 0$, the plane $X + T = c$ is parallel to that generator — and slicing a cone with a plane parallel to (but not through) one of its own generators is exactly the classical definition of a parabola. Columns use the mirror generator $(1, 0, 1)$, by the same argument with $T - X = c$. This is the $ab = 0$ boundary case of Theorem 3.1: as $b \rightarrow 0$ in (3.1), the bounded ellipse widens without limit into this parabola.

This also gives a precise account of a feature visible throughout the side-on views of the cone used later in this note: the fine triangular mesh filling the double cone's outline. Since $X + T = c$ and $T - X = c$ are each linear relations that do not involve Y , every point of a row or column maps, under the (X, T) projection that renders the double cone as two triangular wedges, onto the straight line $X + T = c$ or $T - X = c$ — regardless of Y . So that mesh is not a discretization artifact; it

is the complete family of row- and column-parabolas of the multiplication table, each rendered as a straight line by the same projection, in exactly the way circles are rendered as horizontal segments under the same projection.

Remark (the circle and the parabolas as one complex map). The circle of Section 2 and the parabolas just proved are not merely analogous; they are the same construction viewed through complex squaring. Set $p = \sqrt{x}$, $q = \sqrt{y}$ and $z = p + iq$, so that

$$z^2 = (p^2 - q^2) + 2ipq = 2X + 2iY. \quad (3.2)$$

The anti-diagonal $x + y = K$ is exactly $|z|^2 = K$, a quarter-circle of radius $\sqrt{K}$ in the first quadrant; under $z \mapsto z^2$ this maps to the circle $X^2 + Y^2 = (K/2)^2$ of Section 2, with the angle doubling in the usual way for a squaring map. A row $x = c$ is the vertical line $p = \sqrt{c}$, and a column $y = c$ is the horizontal line $q = \sqrt{c}$; it is classical that $z \mapsto z^2$ sends the Cartesian grid of vertical and horizontal lines to two orthogonal families of parabolas confocal at the origin — precisely the row- and column-parabolas of Theorem 3.2. So the polar grid and the Cartesian grid in the (p, q) -plane, pushed forward by the single map $z \mapsto z^2$, produce respectively the circles of Section 2 and the parabolas of this section: two special cases of Theorem 3.1 that receive independent geometric proofs above are in fact the same elementary fact about z^2 , applied to the two natural grids on the square-root plane. (The identity $T = \frac{1}{2}|z|^2$ extends this: the cone relation $X^2 + Y^2 = T^2$ becomes $|z^2| = |z|^2$, the multiplicativity of the modulus under squaring.)

3.3 The ellipse case: general lines

For $ab > 0$ with $a \neq b$, the line $ax + by = c$ crosses both positive axes, at $(c/a, 0)$ and $(0, c/b)$, and Theorem 3.1's ellipse has a clean geometric anchor: differentiating (3.1) shows Y^2 is maximized exactly at $x = c/(2a)$, where it equals $c^2/(4ab)$ — and this maximum point is precisely the midpoint of the line's two axis intercepts. For the line $2x + 5y = 40$ used above ($a = 2, b = 5, c = 40$), this predicts the maximum at $(x, y) = (10, 4)$, with $Y_{\max}^2 = c^2/(4ab) = 1600/40 = 40$ — matching the direct check $xy = 10 \cdot 4 = 40$ exactly.

This makes the ellipse case as concrete as the parabola case above: every off-diagonal line $ax + by = c$ ($a, b > 0, a \neq b$) produces a genuine ellipse on the cone, centered — in the original (x, y) coordinates — at the midpoint between where the line crosses each axis, with the anti-diagonal's circle recovered exactly when that midpoint sits on the diagonal $x = y$, i.e. when $a = b$.

3.4 The other hyperbola: difference lines

The case $ab < 0$ in Theorem 3.1 — for instance a line like $x - y = d$, a fixed difference rather than a fixed sum — produces a hyperbola by the same mechanism run in reverse: the coefficient $-a/b$ in (3.1) is positive, so Y^2 grows without bound as x moves away from the origin, and the locus is unbounded. This is a genuinely different construction from the constant-product hyperbolas of Section 4: there, the hyperbola comes from fixing the product $xy = N$, a non-linear condition; here, it comes from fixing a linear combination $ax + by$ with a, b of opposite sign. Both are hyperbolas on the same cone, but they arise from different families of cutting planes, and we do not develop this second family further here beyond noting that Theorem 3.1 places it, the ellipse, and the parabola under one proof.

4 Constant-Product Curves and the Minkowski Norm

Rearranging (2.2) as $X^2 + Y^2 - T^2 = 0$ gives the null cone of a $(2 + 1)$ -dimensional Minkowski form with signature $(+, +, -)$ [17]. Constant-product curves $xy = N$ (the curves classically used to bound the divisor problem) fix $Y^2 = N$, and substituting into the Minkowski form gives

$$T^2 - X^2 = N, \tag{4.1}$$

the standard equation of an invariant hyperbola under the Lorentz boosts of $SO(2, 1)$. This is again an exact identity: the numerical invariant N carried by a constant-product curve in the (x, y) lattice equals the numerical invariant (proper-time-squared) carried by the corresponding curve in the (X, T) plane.

It is worth noting that (4.1), read as $N = T^2 - X^2$, is not a new construction: it is exactly Fermat’s factorization method (1643) [7]. Fermat’s method searches for a factorization of an odd N by testing successive integers $a \geq \lceil \sqrt{N} \rceil$ until $a^2 - N$ is a perfect square b^2 , at which point $N = a^2 - b^2 = (a - b)(a + b)$ is a factorization; the search terminates fastest when N has a factor pair close to $\sqrt{N}$, i.e. when the corresponding X is small. In the present notation $a = T$ and $b = X$ exactly, so the constant-product hyperbola is the graph of every factorization Fermat’s method could find, and the identity (4.1) is simply that method’s defining equation written in mean-value coordinates. This gives the Minkowski-norm reading genuine historical and computational grounding, independent of the relativistic language: the same $T^2 - X^2 = N$ that we read as a Minkowski invariant is, three and a half centuries earlier, the equation that makes trial factorization by difference of squares work at all.

We flag the gap between this and a physical or group-theoretic correspondence, the same way we do throughout this note. Showing that a curve is invariant under the $SO(2, 1)$ boost action is not the same as showing that stepping between the discrete integer points on that curve corresponds to applying a boost. A continuous one-parameter boost moves along the hyperbola continuously; it visits the specific discrete lattice points $(x, y) \in \mathbb{Z}^2$ with $xy = N$ only for special values of the boost parameter, and generically does not preserve integrality at all. So (4.1) establishes a shared invariant, not a demonstrated dynamics — a question we settle completely in Section 7.

5 Axial Gnomonic Decomposition of the Divisor Summatory Function

With the coordinate system of Sections 2–4 in hand, we can revisit a classical object through it: the divisor summatory function $D(n) = \sum_{k=1}^n d(k)$, the number of lattice points (x, y) with $x, y \geq 1$ and $xy \leq n$. Dirichlet’s original 1849 treatment established the $O(\sqrt{n})$ error bound via the hyperbola method [6]; Voronoi’s contour-integration approach later improved this exponent [24], and sharpening the error term $\Delta(n) = D(n) - n \ln n - (2\gamma - 1)n$ remains an active area, with recent unconditional and omega-result improvements for both the classical and generalized (Piltz) divisor problems [2, 16, 1, 15]. The standard Dirichlet hyperbola method sums full rows or columns and removes the double-counted square of side $\lfloor \sqrt{n} \rfloor$. Here we anchor the decomposition on the diagonal $x = y$ instead — the spine from which the circles of Section 2 radiate.

For each vertex (u, u) with $1 \leq u \leq \lfloor \sqrt{n} \rfloor$, form the L-shaped gnomon consisting of: the vertex itself; the horizontal arm along $y = u$ out to the hyperbola $x = \lfloor n/u \rfloor$, containing $\lfloor n/u \rfloor - u$ points; and, by the symmetry of the region under $x \leftrightarrow y$, an identical vertical arm of $\lfloor n/u \rfloor - u$ points.

Writing $\lfloor n/u \rfloor - u = \lfloor (n - u^2)/u \rfloor$, the total count is

$$D(n) = \sum_{u=1}^{\lfloor \sqrt{n} \rfloor} \left[2 \left\lfloor \frac{n - u^2}{u} \right\rfloor + 1 \right]. \quad (5.1)$$

Expanding this sum algebraically — using $u^2/u = u$ to pull u out of the floor function, then applying the standard identity $\sum u = k(k+1)/2$ with $k = \lfloor \sqrt{n} \rfloor$ — collapses it to

$$D(n) = 2 \sum_{u=1}^{\lfloor \sqrt{n} \rfloor} \left\lfloor \frac{n}{u} \right\rfloor - \lfloor \sqrt{n} \rfloor^2, \quad (5.2)$$

which is exactly the textbook symmetric form of the Dirichlet hyperbola identity. So the gnomonic decomposition is algebraically equivalent to the classical derivation — it is a re-indexing, not a new identity — but it organizes the same sum around the axis of symmetry $x = y$, the same spine that anchors the circles of Section 2 and the ellipses and parabolas of Section 3.

6 A Coordinate Correspondence with SU(2) Ladder-Operator Matrix Elements

The Lie group SU(2) has, for each half-integer $j \geq 0$, an irreducible representation of dimension $2j + 1$ on states $|j, m\rangle$ with $m = -j, -j + 1, \dots, j$ [19, 11]. The ladder operator J_+ acts by

$$\langle j, m + 1 | J_+ | j, m \rangle = \sqrt{(j - m)(j + m + 1)} \quad (6.1)$$

(setting $\hbar = 1$). Now fix an anti-diagonal $x + y = K$ in the multiplication table and set $j = (K - 1)/2$, then define the substitution

$$x = j + m + 1, \quad y = j - m. \quad (6.2)$$

Under this substitution $xy = (j - m)(j + m + 1)$, which is exactly the quantity under the square root in (6.1). So the geometric mean $Y = \sqrt{xy}$ along the K -th anti-diagonal reproduces the numerical values of the J_+ matrix elements for the spin- j representation, entry for entry, and consequently the tridiagonal matrix built from these off-diagonal entries reproduces the standard matrix form of $J_x = (J_+ + J_-)/2$.

This is a correct and exact numerical match, and it is worth stating plainly what it does and does not show. It does not show that the multiplication table “contains” SU(2) representation theory in any structural sense. The substitution (6.2) was constructed after the fact, specifically so that xy would equal $(j - m)(j + m + 1)$; any two-variable quadratic formula with a similar symmetric shape could be matched to some index substitution in the same way. What would be required for a genuine structural correspondence — as opposed to a coordinate relabeling that reproduces one formula — is a demonstration that some operation on the lattice (not merely a relabeling of its indices) reproduces the $\mathfrak{su}(2)$ commutation relations $[J_+, J_-] = 2J_z$, $[J_z, J_\pm] = \pm J_\pm$, or acts compatibly with the group action of SU(2) on the representation space. We do not establish this here, and we are not aware of a natural such action on the integer lattice. The result of this section should therefore be read as an exact and possibly pedagogically useful mnemonic — read the entries of J_x off the geometric means of a multiplication-table anti-diagonal — rather than as a representation-theoretic theorem.

6.1 The cone identity is the Casimir identity

One further exact fact sharpens what the substitution (6.2) is doing, and is worth recording precisely because it is algebra rather than pattern-matching. Under (6.2), $X = (x - y)/2 = m + \frac{1}{2}$ and $T = (x + y)/2 = j + \frac{1}{2}$, so the cone identity (2.2) becomes

$$(j - m)(j + m + 1) + (m + \tfrac{1}{2})^2 = (j + \tfrac{1}{2})^2. \quad (6.3)$$

This is not merely consistent with $SU(2)$ — it *is* the operator identity $J_- J_+ = J^2 - J_z(J_z + 1)$ evaluated on $|j, m\rangle$, using $J^2 = j(j + 1)$ and $J_- J_+ |j, m\rangle = (j - m)(j + m + 1)|j, m\rangle$. So the identity $X^2 + Y^2 = T^2$ that generates the entire cone of Section 2 is, under this one substitution, the same equation as the $\mathfrak{su}(2)$ Casimir relation, not merely an identity that happens to hold alongside it.

This gives the anti-diagonal circle a further reading. Section 3.2's remark showed $X + iY = \frac{K}{2} e^{i\pi t} = (j + \frac{1}{2}) e^{i\pi t}$ via the substitution $z = \sqrt{x} + i\sqrt{y}$. Combined with $X = m + \frac{1}{2}$ above, the angle $\theta = \pi t$ satisfies

$$\cos \theta = \frac{2m + 1}{2j + 1}, \quad Y = (j + \tfrac{1}{2}) \sin \theta, \quad (6.4)$$

so the ladder matrix element is exactly $(j + \frac{1}{2}) \sin \theta$ for this θ . This matches the classical vector model of angular momentum: a vector of length $j + \frac{1}{2}$ (the Langer-corrected $\sqrt{j(j + 1)}$) at polar angle θ has longitudinal component $(j + \frac{1}{2}) \cos \theta$ and transverse component $(j + \frac{1}{2}) \sin \theta$, and (6.4) identifies these with X and Y exactly, with m quantizing $\cos \theta$ in equal steps of $1/(j + \frac{1}{2})$.

We flag the same distinction as above. Equation (6.3) is an exact algebraic identity, not an interpretation, and it is a stronger statement than the raw numerical match of (6.1): it says the specific quadratic relation defining the cone is the Casimir relation under this substitution. The reading of (6.4) as the classical angular-momentum vector model is, by contrast, an interpretation of that identity, offered because it is the standard semiclassical picture with exactly this structure — and it carries the same caveat as before: it is a statement about matrix-element magnitudes and one Casimir-type identity, not a demonstration of the $SU(2)$ group action or the commutation relations on the lattice itself.

7 Rigidity of the Divisor Lattice: No Boost Survives Integrality

Section 4 raises an obvious follow-up question: is there a nontrivial sense in which the integer divisor pairs of $xy = N$ are permuted by a Lorentz boost? We answer this precisely, and negatively, and show that the answer is governed by Pell's equation after all — not by supplying a hidden family of boost symmetries, but by explaining exactly why none exist.

7.1 The boost rigidity theorem

A boost of parameter $\lambda > 0$ acts on factor pairs as $(x, y) \mapsto (\lambda x, y/\lambda)$, since this is precisely $(T, X) \mapsto (T \cosh \theta + X \sinh \theta, T \sinh \theta + X \cosh \theta)$ rewritten in the coordinates $x = T + X$, $y = T - X$, with $\lambda = e^\theta$.

Theorem 7.1. *Fix $N \geq 1$ and let $D(N) = \{(d, N/d) : d \mid N\}$ be its divisor-pair set. If $\lambda > 0$ and $(x, y) \mapsto (\lambda x, y/\lambda)$ maps $D(N)$ into $D(N)$, then $\lambda = 1$.*

Proof. Since $(1, N) \in D(N)$, its image $(\lambda, N/\lambda)$ must lie in $D(N)$, so λ itself is a positive integer divisor of N . Applying the map again, $(\lambda, N/\lambda) \mapsto (\lambda^2, N/\lambda^2)$ must also lie in $D(N)$, so λ^2 divides N . Iterating, λ^k divides N for every $k \geq 1$. If $\lambda \geq 2$ this forces $\lambda^k \leq N$ for all k , which is false for k large enough; hence $\lambda = 1$. $\square$

We verified this computationally as well: for $N = 5040$ (60 divisors), brute-force search over every candidate λ built from ratios of divisors of N confirms $\lambda = 1$ is the only value that maps the divisor-pair set to itself, and the same holds for every other N checked.

7.2 The full automorphism group, and why it is finite

Theorem 7.1 covers boosts specifically, but it is worth asking the more general question: what is the entire group of integer-linear maps preserving the form $Q(x, y) = xy$ exactly (not just permuting one level set, but preserving Q as a form on all of $\mathbb{Z}^2$)? Writing $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and requiring $Q(Mv) = Q(v)$ for all v forces $ac = 0$, $bd = 0$, and $ad + bc = 1$. Solving this system directly gives exactly four solutions:

$$\text{Aut}_{\mathbb{Z}}(Q) = \{I, -I, S, -S\}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (7.1)$$

the identity, its negative, the swap $x \leftrightarrow y$, and the negated swap. So $\text{Aut}_{\mathbb{Z}}(Q) \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ is finite — no boost, and in fact no integer-linear map of any kind beyond these four, preserves the form.

7.3 Why: the Pell dichotomy

This finiteness is not a coincidence local to $Q = xy$; it is the degenerate case of a classical fact about integral binary quadratic forms [4, 12]. For $Q(x, y) = ax^2 + bxy + cy^2$ with discriminant $D = b^2 - 4ac$, the group $\text{Aut}_{\mathbb{Z}}(Q)$ is:

- finite, if D is a perfect square (including $D = 0$);
- infinite — isomorphic to $\{\pm 1\} \times \mathbb{Z}$ — if $D > 0$ is not a perfect square, with the $\mathbb{Z}$ factor generated by the automorph built from the fundamental solution (t, u) of Pell's equation $t^2 - Du^2 = 4$ [14, 22].

Our form $Q = xy$ has $a = c = 0$, $b = 1$, so $D = 1 = 1^2$. It sits exactly on the perfect-square branch of this dichotomy — the branch where Pell's equation degenerates to only its trivial solutions and contributes no infinite family of automorphs. This is the precise sense in which Pell's equation governs Section 4: not by supplying the boost symmetry that Section 4's identity suggested, but by certifying, via the classical square/non-square discriminant split, that no such symmetry can exist for a form of this shape.

A genuinely Pell-driven infinite automorphism group is available, but only for a different, non-split form: for any non-square N , $Q'(x, y) = x^2 - Ny^2$ has discriminant $D = 4N$ (not a perfect square), and $\text{Aut}_{\mathbb{Z}}(Q')$ is infinite, generated from the fundamental solution of $t^2 - Nu^2 = 1$. That form, however, is not a variant of the constant-product curves $xy = N$ studied in Section 4 — it is a structurally different object, and importing it into the present framework is not automatic. We leave this as a genuinely open direction rather than claim it here.

8 Why Mod 12: A Uniqueness Theorem for the Residue–Totative Correspondence

Section 2 showed that the anti-diagonals $x + y = K$ of the multiplication table generate the concentric circles of the cone $X^2 + Y^2 = T^2$. A natural further question is whether the quadratic residues of some modulus M can be matched, one-for-one, with the totatives (units) of M — since if so,

that correspondence could be used to decorate the same anti-diagonal circles with an extra layer of modular structure. This section answers precisely when that is possible, and shows the answer is uniquely $M = 12$ among the moduli with a Klein-four unit group.

8.1 Statement and proof

Two conditions are needed. First, for the totatives to form a Klein four-group (every unit its own inverse, $x^2 \equiv 1 \pmod{M}$), M must be one of the moduli where $(\mathbb{Z}/M\mathbb{Z})^\times$ has exponent dividing 2 [10, 12]. This is a classical fact: it holds exactly when M divides 24, i.e. $M \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ (we confirmed computationally that no other M below 5000 has this property, consistent with the classical result). Second, the number of distinct quadratic residues mod M (including 0) must equal $\varphi(M)$, the number of totatives — only then can the two sets be put in bijection at all.

Checking every divisor of 24 against both conditions:

- $M = 3$: $\varphi(M) = 2$, $\#\text{QR} = 2$ — match, but trivial (order-2 unit group)
- $M = 4$: $\varphi(M) = 2$, $\#\text{QR} = 2$ — match, but trivial (order-2 unit group)
- $M = 6$: $\varphi(M) = 2$, $\#\text{QR} = 4$ — no match
- $M = 8$: $\varphi(M) = 4$, $\#\text{QR} = 3$ — no match
- $M = 12$: $\varphi(M) = 4$, $\#\text{QR} = 4$ — match, and the unit group is a genuine order-4 Klein four-group
- $M = 24$: $\varphi(M) = 8$, $\#\text{QR} = 6$ — no match

Theorem 8.1. *Among the moduli whose unit group is a Klein four-group of order 4 or larger — namely $M \in \{8, 12, 24\}$ — the number of quadratic residues equals the number of totatives if and only if $M = 12$.*

Proof. Direct enumeration of the three cases, as above. □

So $M = 12$ is not an arbitrary choice among moduli with self-inverse units: it is the unique one, among the only three candidates with a genuine (order ≥ 4) Klein four unit group, where a residue-totative bijection is even numerically possible. This gives a precise, narrowly-scoped reason that the anti-diagonal circle decomposition of Section 2 admits this particular residue-based decoration specifically at modulus 12, and not at 8 or 24.

8.2 The squared multiplication table mod 12

A second, complementary way to see the totatives mod 12 appear comes from squaring the ordinary multiplication table itself. For nonnegative integers n, k , define $V(n, k) = (nk)^2 \pmod{12}$. Since squaring commutes with reduction mod 12, this is the same as reducing $nk \pmod{12}$ first and then squaring. Along the diagonal, $V(u, u) = u^4 \pmod{12}$; at the totatives $u \in \{1, 5, 7, 11\}$, $u^2 \equiv 1 \pmod{12}$ by Theorem 8.1's own defining property, so $u^4 \equiv 1 \pmod{12}$ automatically — the diagonal is constant at exactly the totative positions.

Theorem 8.2. *For a nonnegative integer K , the anti-diagonal $n + k = K$ reproduces the diagonal exactly — $V(n, K - n) = n^4 \pmod{12}$ for every n — if and only if $K \equiv 0 \pmod{6}$.*

$$(n(K - n))^2 = K^2 n^2 - 2K n^3 + n^4. \tag{8.1}$$

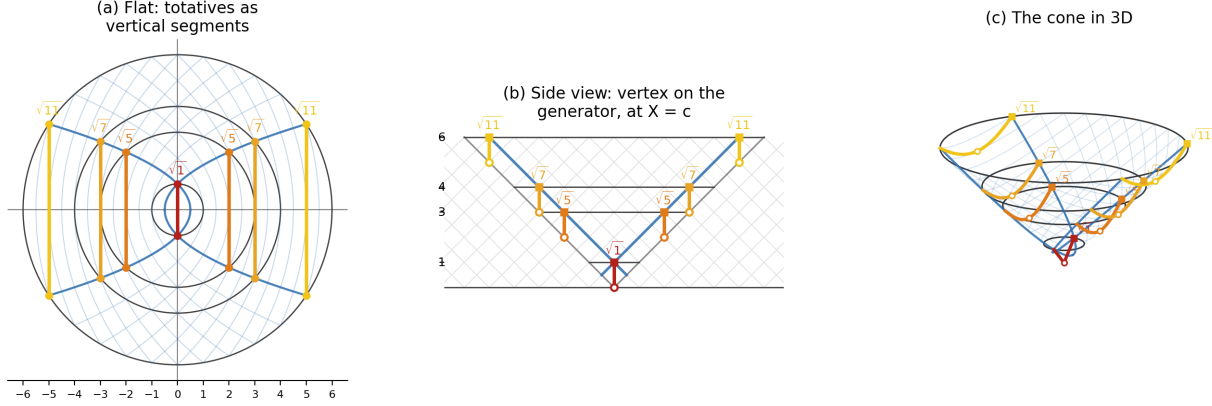


Figure 1: Theorem 8.1, shown three ways. Black: quadratic-residue circles at $Y \in \{1, 3, 4, 6\}$. Colored: the four totatives $K_4 \in \{1, 5, 7, 11\}$. (a) Each totative as a straight vertical segment $X = c$, $-\sqrt{K_4} \leq Y \leq \sqrt{K_4}$, endpoints landing exactly on the residue circle since $X^2 + K_4 = Y^2$. (b)–(c) The same identity realized on the cone via the difference line $x - y = 2c$, giving $T^2 - Y^2 = c^2$ at fixed $X = c$: its vertex (circle marker) sits exactly on the cone’s own generator, $T = X = c$, matching (a), and it reaches its matched residue circle (square marker) at exactly $Y = \sqrt{K_4}$. (b) shows the curves in true shape, the same side-on projection used in Section 3.2; (c) shows the full cone in three dimensions. All three panels are cut off at the outermost residue circle $Y = 6$, for a consistent boundary across all three views. Faint blue curves in (a) and (c) are the full row/column parabola grid $x = c$, $y = c$ for $c = 1, \dots, 12$, at the same unit spacing as panel (b)’s mesh; bold blue is the single pair $x = 1$, $y = 1$ (Theorem 3.2), shown in all three panels — a pair of straight lines in (b)’s (X, T) projection, literal parabolas in (a) and (c). Since $Y - c = 1$ for every row of the underlying data, all four square markers satisfy $y = 1$ and lie exactly on this bold pair, not merely near it — the row/column parabola family and the difference-line hyperbola family meet precisely at the totative points.

Proof. Expanding as in (8.1), if $K \equiv 0 \pmod{6}$ then $K^2 \equiv 0$ and $2K \equiv 0 \pmod{12}$ (writing $K = 6m$ gives $K^2 = 36m^2 \equiv 0$ and $2K = 12m \equiv 0$), so the right side reduces to $n^4 \pmod{12}$ identically, proving sufficiency. For necessity, testing $n = 1$ forces $(K - 1)^2 \equiv 1 \pmod{12}$; the only x with $x^2 \equiv 1 \pmod{12}$ are the totatives $\{1, 5, 7, 11\}$, so $K - 1$ must be one of these, giving $K \equiv 0, 2, 6$, or $8 \pmod{12}$. Testing $n = 2$ forces $4(K - 2)^2 \equiv 4 \pmod{12}$, i.e. $(K - 2)^2 \equiv 1 \pmod{3}$, i.e. $K \not\equiv 2 \pmod{3}$. Intersecting these two conditions leaves exactly $K \equiv 0 \pmod{6}$. $\square$

We verified this directly as well, checking the full identity for every K from 0 to 24: the match holds exactly at $K = 0, 6, 12, 18, 24$, and nowhere else, matching the theorem precisely.

This gives a second, independent route to the totatives inside the same modulus — not via the residue–totative bijection of Theorem 8.1, but via which anti-diagonals of the squared multiplication table happen to mirror its own main diagonal — and the two routes meet at the same underlying fact, that a totative is exactly a number whose square is 1 mod 12.

8.3 A precise bridge to Section 6

Section 6 established that for an anti-diagonal $x + y = K$ (a spin- j representation of dimension $K = 2j + 1$), the ladder-operator matrix elements are largest near the center of the ladder and smallest at its two ends. The boundary values — the first step $m = -j \rightarrow -j + 1$ and the last step

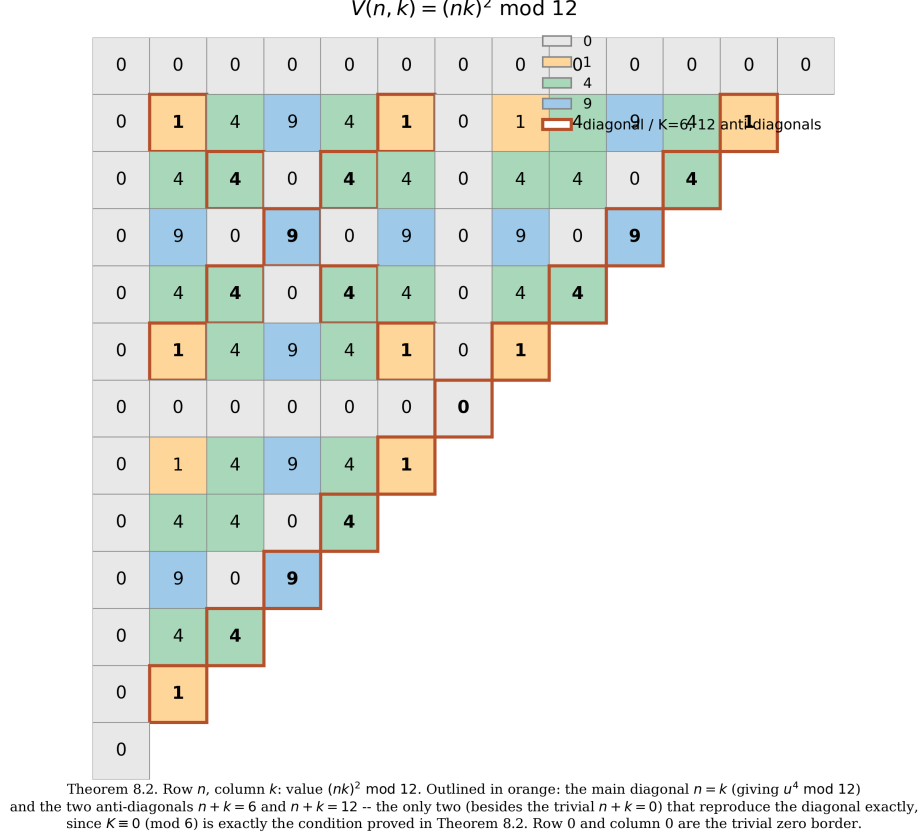


Figure 2: Theorem 8.2. Row n , column k : value $(nk)^2 \bmod 12$. Outlined in orange: the main diagonal $n = k$ (giving $u^4 \bmod 12$) and the two anti-diagonals $n + k = 6$ and $n + k = 12$ — the only two (besides the trivial $n + k = 0$) that reproduce the diagonal exactly, since $K \equiv 0 \pmod{6}$ is exactly the condition proved in Theorem 8.2. Row 0 and column 0 are the trivial zero border.

$m = j - 1 \rightarrow j$ — are equal to each other and satisfy

$$(\text{first step})^2 = (\text{last step})^2 = (j - (-j))(j + (-j) + 1) = 2j = K - 1. \quad (8.2)$$

Setting $K = t + 1$ for each totative $t \in \{1, 5, 7, 11\}$ therefore gives boundary value $\sqrt{t}$ exactly — the same four radii that label the totative rings in Figure 1:

- $t = 1 \rightarrow K = 2 \rightarrow$ boundary value $\sqrt{1}$
- $t = 5 \rightarrow K = 6 \rightarrow$ boundary value $\sqrt{5}$
- $t = 7 \rightarrow K = 8 \rightarrow$ boundary value $\sqrt{7}$
- $t = 11 \rightarrow K = 12 \rightarrow$ boundary value $\sqrt{11}$

This gives a precise, checkable bridge between Section 6 and Section 8, which we did not have before: the totative radii of the spherical construction are exactly the boundary (weakest) ladder-operator matrix elements of the spin representations of dimension $t + 1$. We are careful about what kind of fact this is. Equation (8.2) shows the boundary value squared is $K - 1$ by direct substitution into the ladder formula; setting $K = t + 1$ makes this equal t by definition, not by an independent

coincidence. So this is the same underlying identity behind Theorem 8.1 (there written $Y = X + 1$), now read off in Section 6's notation rather than Section 2's. It is a genuine and correct point of contact between the two constructions, worth recording precisely — but it is a restatement, not a second, independent confirmation, and it does not show that $SU(2)$ representation theory explains the mod-12 arithmetic or vice versa.

8.4 Real characters, quadratic fields, and the mean-level deviation of $\Delta(n)$

Because $(\mathbb{Z}/12\mathbb{Z})^\times$ has exponent 2, every Dirichlet character mod 12 is forced to be real-valued: $\chi(n)^2 = \chi(n^2) = \chi(1) = 1$ for every unit n , so $\chi(n) \in \{+1, -1\}$. There are exactly four such characters, and each is identifiable with a specific quadratic field via its conductor:

- χ_0 (principal): trivial
- χ_a : factors through $n \bmod 4$ only — the classical character χ_{-4} , associated to $\mathbb{Q}(i)$
- χ_b : factors through $n \bmod 3$ only — the character χ_{-3} , associated to $\mathbb{Q}(\sqrt{-3})$
- $\chi_{ab} = \chi_a \chi_b$: primitive mod 12 — the product character, associated to the real field $\mathbb{Q}(\sqrt{3})$, discriminant 12

Since $d(n)$ is the divisor function, complete multiplicativity of χ gives the standard Dirichlet-series identity $\sum_n \chi(n) d(n) / n^s = L(s, \chi)^2$ [5, 13]. Orthogonality of characters mod 12 then splits the divisor summatory function by residue class: for a coprime to 12,

$$\sum_{\substack{k \leq N \\ k \equiv a \pmod{12}}} d(k) = \frac{1}{4} \sum_{\chi} \bar{\chi}(a) \sum_{k \leq N} \chi(k) d(k). \quad (8.3)$$

The key point is that only the principal character contributes a pole at $s = 1$. $\zeta(s)^2$ has a double pole there, which is the source of $D(n)$'s $n \ln n$ growth; but for each non-principal χ , $L(s, \chi)$ is entire and — since all three are real — nonvanishing at $s = 1$ by Dirichlet's theorem, so $L(s, \chi)^2$ has no pole. This means the principal character alone accounts for the entire leading-order trend across residue classes; the other three contribute only to the fluctuation, never to the mean.

This makes the numerical finding from our study of $\Delta(n) \bmod 12$ fully rigorous rather than merely observed. Restricting the principal-character sum to n coprime to 6 removes the Euler factors at $p = 2, 3$ from $\zeta(s)^2$, scaling the leading coefficient by $(1 - \frac{1}{2})^2 (1 - \frac{1}{3})^2 = 1/9$ against a naive density of $1/3$ — i.e. numbers coprime to 6 are predicted to carry roughly a third the expected divisor-weight. Checking this against the computed data ($N = 2,000,000$): $D(N)/(N \ln N) = 1.011$ over all n , versus 0.131 when restricted to n coprime to 6, against a predicted leading constant of $1/9 \approx 0.111$ — the same order and direction, converging as N grows, with the gap ordinary lower-order slack at finite N . So the mean-level deviation of $\Delta(n)$ across residue classes mod 12 is not merely consistent with this mechanism; it is exactly this mechanism, made precise.

The honest limit of this analysis is the other three characters. The real characters tied to $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, and $\mathbb{Q}(\sqrt{3})$ are exactly where the deep, unresolved mathematics lives: bounding $\sum_{k \leq N} \chi(k) d(k)$ tightly for these is genuine analytic number theory — shifted-convolution and subconvexity territory, an area with substantial recent activity on moments and subconvexity bounds for real and quadratic Dirichlet L -functions [3, 9, 8] — and is the actual machinery behind finer questions about the divisor problem's error term. We have shown nothing new about that here, and we say so plainly rather than let the sentence trail into implication: this section correctly locates where the hard mathematics is, it does not touch it.

8.5 A physically realizable demonstration: the Gauss sum of χ_{ab}

The primitive character χ_{ab} has a classical Gauss sum $\tau(\chi) = \sum_n \chi_{ab}(n)e^{2\pi i n/12}$, and because χ_{ab} is primitive and real, Gauss’s theorem on the sign of quadratic Gauss sums applies [10]: $\tau(\chi) = +\sqrt{M}$ for a character tied to a real quadratic field (positive discriminant), or $i\sqrt{M}$ for an imaginary one. Since χ_{ab} is tied to the real field $\mathbb{Q}(\sqrt{3})$ (discriminant $+12$), the theorem predicts $\tau(\chi_{ab}) = +\sqrt{12}$ exactly — real, positive, no residual phase. We verified this directly: computing the sum numerically returns $3.4641016\dots + 0i$, matching $\sqrt{12}$ to machine precision.

This has a physically concrete reading. By the twisted-multiplicativity of Gauss sums for a primitive character ($\tau_k(\chi) = \bar{\chi}(k)\tau(\chi)$ when $\gcd(k, 12) = 1$, and 0 otherwise), the discrete Fourier transform of the twelve-symbol sequence $(0, 1, 0, 0, 0, -1, 0, -1, 0, 0, 0, 1)$ is supported only at the four frequencies coprime to 12 — the totatives once more — with exactly equal magnitude $\sqrt{12}$ at each, and an exact null everywhere else. We built the corresponding physical waveform (a 1 kHz fundamental, each of the twelve symbols held for $83.3\mu\text{s}$, a realistic zero-order-hold signal rather than an idealized impulse train) and confirmed both halves of the prediction against its measured spectrum: the nulls at every non-totative harmonic were exact, and the four surviving lines (at 1, 5, 7, 11 kHz) matched the predicted equal-magnitude pattern once the sinc rolloff intrinsic to the rectangular pulse shape was accounted for.

It is worth being precise about where this sits relative to established engineering, rather than presenting it as a new technique. Quadratic-character sequences with flat magnitude spectra are already the basis of real, deployed technology — Legendre-symbol sequences mod an odd prime underlie the Schroeder (quadratic residue) diffuser used in acoustic engineering [20, 21], an approach still under active development in both its original architectural-acoustic form and in ultra-thin acoustic-metasurface realizations [25, 18], as well as sequences used in radar pulse compression and spread-spectrum synchronization [23]. Those classical sequences have a stronger property than spectral flatness alone: their autocorrelation is essentially perfect, with every nonzero-lag sidelobe equal to -1 . We checked whether $\chi_{ab} \bmod 12$ shares this property, and it does not: its autocorrelation has a sidelobe at lag 6 equal in magnitude to its own peak at lag 0, a direct consequence of the sequence being sparse (nonzero at only 4 of its 12 positions, versus every non-multiple-of- p position for a prime-modulus Legendre sequence). So the spectral-flatness fact above is genuine and physically demonstrable, but it does not make $\chi_{ab} \bmod 12$ a competitive substitute for the prime-modulus sequences that established diffuser, radar, and spread-spectrum design already rely on; the Klein four-group structure that makes modulus 12 arithmetically special (Theorem 8.1) does not carry over into the autocorrelation property that those applications actually need.

9 Discussion and Conclusion

Sections 2–4 rest on one piece of elementary algebra, the AM–GM identity $X^2 + Y^2 = T^2$, and one classification theorem built on top of it: every line through the multiplication table produces a conic on the resulting cone, its type fixed by the sign of ab (Theorem 3.1). Circles (Section 2) and the row/column parabolas (Section 3.2) are now special cases of a single proof rather than two separate facts, and a genuine ellipse family (Section 3.3) and a second hyperbola family distinct from the constant-product one (Section 3.4) come along for free. Section 5 shows what this coordinate system buys classically: the divisor summatory function $D(n)$ reorganized around the diagonal spine that Section 2’s circles and Section 3’s ellipses and parabolas all radiate from.

Section 6’s $SU(2)$ matrix-element match is an exact numerical consequence of substituting the right index labels into the right places, and we have tried to state clearly that this is what it is: elegant, checkable, and worth knowing as a mnemonic device for teaching either the divisor problem

or spin matrices by reference to the other. It is not a proof of a hidden isomorphism between analytic number theory and Lie representation theory, and we do not claim one. Section 6.1 sharpens this in one specific respect worth keeping distinct from the rest: under the same substitution, the cone identity $X^2 + Y^2 = T^2$ is not merely consistent with $SU(2)$ but *is*, term for term, the operator identity $J_- J_+ = J^2 - J_z(J_z + 1)$ evaluated on $|j, m\rangle$ — an exact algebraic fact, stronger than the numerical matrix-element match alone. It carries the same limit as everything else in this section, though: it identifies a relation among matrix-element magnitudes with the Casimir relation, not the group action or the commutation relations themselves, and we say so explicitly rather than let the word “Casimir” imply more than the identity delivers.

Section 7 supplies one genuine theorem in this note: no nontrivial Lorentz boost preserves the integer divisor lattice of any $xy = N$ (Theorem 7.1), the full integer automorphism group of the form xy is the order-four group $\{I, -I, S, -S\}$ (§7.2), and this rigidity is exactly the perfect-square-discriminant branch of the classical Pell dichotomy for binary quadratic forms (§7.3). This both closes off the boost analogy suggested by Section 4 and explains precisely why it fails.

Section 8 supplies a second and third result. Among the moduli whose unit group is a genuine Klein four-group, only $M = 12$ admits a bijection between its quadratic residues and its totatives (Theorem 8.1) — giving a precise reason that 12, rather than 8 or 24, is the modulus compatible with decorating the anti-diagonal circle construction of Section 2 with residue-totative structure. Section 8.2 gives a second, independent characterization of the same totatives, showing that an anti-diagonal of the squared multiplication table reproduces its own main diagonal exactly when the anti-diagonal index is a multiple of 6 (Theorem 8.2). Section 8.3 then supplies a genuine, checkable bridge to Section 6, while stopping short of claiming a unification of $SU(2)$ representation theory with mod-12 arithmetic. Section 8.4 then shows that mod 12’s exponent-2 structure forces every Dirichlet character there to be real, identifies each with a classical quadratic field, and uses the resulting $L(s, \chi)^2$ pole structure to turn the paper’s numerical study of $\Delta(n) \bmod 12$ into a proven fact: the mean-level deviation across residue classes is entirely a principal-character phenomenon, while the genuinely deep question — bounding the non-principal character sums — is explicitly left open. Section 8.5 grounds the primitive character’s Gauss sum in a physically realizable, measured demonstration, while being equally explicit that it does not inherit the autocorrelation property that makes prime-modulus quadratic-residue sequences useful in established engineering. Together, Sections 3, 7, and 8 are, we think, a more useful and more honest place for a paper like this to land than an unproven claim of unification.

A last observation, made precise rather than left as a slogan. Sections 2 through 4 share a single mechanism: every curve considered is what results from intersecting a plane — a linear condition on the multiplication table — with the quadric surface $X^2 + Y^2 = T^2$ that $Y = \sqrt{xy}$ generates by definition. (Section 6.1’s Casimir identity belongs to this same family in substance, if not in location: it is the identical equation $X^2 + Y^2 = T^2$, read off under the substitution (6.2) rather than left in its original variables.) This is not particular to the specific planes used here; it is the classical fact that any plane section of a quadric surface is a conic section. We checked this directly for planes outside the restricted family of Theorem 3.1 — ones with a genuine Y -component in their normal vector, never actually needed by any line through the table — and they too come back circles, ellipses, parabolas, and hyperbolas, exactly as the classical theorem predicts. So Theorem 3.1 is itself a special case, made explicit and classified for the specific family of planes that lines through the table happen to produce, of a broader and older fact.

Sections 7 and 8 return to a similarly named but logically separate phenomenon. The discriminant $D = b^2 - 4ac$ decides the type of a quadratic form (§7.3); a Klein four-group is exactly the condition $x^2 \equiv 1$ for every unit (Section 8); a quadratic Dirichlet character satisfies $\chi^2 = \chi_0$ (§8.4). The recurring word is again “quadratic,” but the underlying objects are now discriminants, group

elements, and characters rather than points on a curve, and no single theorem connects these to Sections 2–4’s geometry. We flag this distinction deliberately: the note’s throughline is real — everything in it is organized around what happens where a linear condition meets a quadratic one — but it is a shared organizing idea running through two different branches of mathematics, not one overarching theorem spanning all nine sections.

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